

FLASHEATS - ONLINE CANTEN MANAGEMENT SYSTEM

A MINI PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

Under the supervision of

Dr. Abhishek Nagar & Er. Kajal



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SESSION 2025-2026

CERTIFICATE

This is to certify that **Prajit Kumar Singh, Parth Mishra, Priyanshu, Sachlang Debbarma** and **Umang Rana**, students of the Computer Science and Engineering branch, have successfully completed the mini-project titled “FlashEats: Online Canteen Management Sysytem” during the 4th semester of the B.Tech. program for the academic year 2025– 26.

The project was carried out under the guidance of Dr. Abhishek Nagar and Er. Kajal, during which the students demonstrated commendable diligence, dedication, and commitment.

This project was submitted in partial fulfillment of the requirements of the B.Tech. program and has been duly evaluated by the undersigned.

We extend our best wishes to the students for continued success in future endeavours.

Date: 21/04/2026

Signature of Supervisor

Dr. Abhishek Nagar

Er. Kajal

CSED, IET LKO

FlashEats : Online Canteen Management System

Abstract

FlashEats is a web-based online canteen management system designed to streamline the process of food ordering and order management within a campus environment. Traditional canteen ordering systems are often manual, time-consuming, and lack proper tracking, which leads to long queues, order delays, and inefficient communication between students and canteen staff.

The proposed system provides a digital platform where students can securely register, log in, browse the menu, and place food orders. The system includes role-based access control, allowing administrators to efficiently manage menu items, monitor incoming orders, and update order statuses in real time.

FlashEats is developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) and incorporates modern features such as JWT-based authentication and OTP verification using SendGrid for secure email-based verification. The application ensures secure role-based authorization for different users such as students and administrators.

The system is designed to improve efficiency, reduce waiting time, and enhance the overall user experience in campus food ordering by digitizing the entire canteen workflow without relying on traditional manual processes.

ACKNOWLEDGEMENT

Apart from our efforts, the success of this project depends largely on the encouragement and guidance of many others. We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project.

We would like to express our deep and sincere gratitude to our supervisors, DR. ABHISHEK NAGAR & ER. KAJAL, who gave us his full support and encouraged us to work on an innovative and challenging project in the educational field. His wide knowledge and logical thinking guided us in the right direction at all times. We are also deeply grateful to our project coordinator for the help and support provided at every step of this project.

Last but not least, we thank all our friends and family for their constant support and cooperation.

INDEX OF CONTENTS

01. Introduction

1.1 Key features

02. Survey

03. Requirement and feasibility study

04. Implementation

4.1 Technologies and tools used in project.

4.2 Architecture of the App.

4.3 Core functionalities of the App .

05. Result

5.1 Home Page

5.2 Student Dashboard

5.3 Admin Panel & Complaints Form

5.4 Complaints Management

06. Future Enhancements

07. References

1.INTRODUCTION

In many educational institutions, canteen management still relies on traditional methods such as manual ordering at counters or verbal communication between students and staff. These approaches are often inefficient, lack proper tracking, and frequently result in long waiting times, order mismatches, and delayed service.

FlashEats aims to digitize the canteen ordering process by providing an online platform where students can easily browse the menu, place food orders, and track their order status in real time. At the same time, canteen administrators can efficiently manage menu items, monitor incoming orders, and update order statuses through a centralized dashboard.

This system improves transparency, reduces waiting time, and ensures better coordination and efficiency in campus canteen management.

1.1 Key Features

Role-Based Access Control (RBAC):

Separate dashboards and permissions for students and administrators, secured using JWT-based authentication to ensure controlled access to system functionalities.

Menu Management Module:

Administrators can add, update, and manage menu items with details such as name, category, price, and availability for efficient canteen operations.

Order Management System:

Students can browse menu items and place orders, while administrators can view, manage, and update order statuses for smooth processing.

OTP-Based Email Verification:

Users must verify their email during registration using OTP sent via SendGrid, ensuring authenticity and preventing fake accounts.

Menu Filtering & Search System:

Menu items can be filtered based on category, price, or availability, allowing students to quickly find desired food items.

Real-Time Order Tracking:

Students can track the status of their orders dynamically (e.g., placed, preparing, ready), ensuring transparency in the ordering process.

Dashboard System (Student & Admin Panels):

Provides separate and interactive dashboards where students can place orders and admins can manage menu items and incoming orders efficiently.

Secure Authentication System:

Implements JWT-based authentication along with password hashing to ensure secure login and protection of user data.

Cloud Deployment (AWS EC2 & S3):

Backend services are hosted on AWS EC2, while static assets are managed using AWS S3, ensuring scalability and high availability.

RESTful API Integration:

Backend functionalities are exposed through REST APIs for seamless communication between frontend and backend components.

Responsive & User-Friendly Interface:

Built using React.js and Tailwind CSS to provide a clean, fast, and responsive experience across mobile, tablet, and desktop devices.

2. LITERATURE SURVEY

1. Digital Systems and Operational Automation:

Research highlights that traditional canteen management systems relying on manual ordering often lead to inefficiencies, long waiting times, and order mismatches. Studies show that digital ordering systems significantly improve operational speed, accuracy, and user satisfaction by automating the entire workflow.

2. User Experience and Interface Design:

Scholarly works emphasize that a simple, intuitive, and responsive user interface improves usability and encourages users to interact more effectively with the system. Research suggests that minimal input design, clear navigation, and mobile-friendly layouts significantly enhance the food ordering experience.

3. MERN Stack and Web Application Development:

Technical literature demonstrates that the MERN stack (MongoDB, Express.js, React.js, Node.js) enables the development of scalable and high-performance web applications. React ensures smooth and dynamic UI rendering, while Node.js and Express provide efficient backend processing and API handling for real-time systems like FlashEats.

4. Authentication and Security Mechanisms:

Literature emphasizes the importance of secure authentication in modern web applications to protect user data. Studies recommend JWT-based authentication for session management and OTP-based email verification using services like SendGrid to ensure secure and verified user access.

5. Cloud Deployment and Scalability:

Industry reports indicate that cloud platforms such as AWS enhance application reliability, scalability, and performance. Services like AWS EC2 for backend hosting and AWS S3 for static storage ensure high availability and efficient resource management for web-based systems.

6. Industry Trends and User Expectations:

Current trends show a growing demand for digital food ordering platforms that offer real-time order tracking, fast service, and seamless user experience. User feedback from similar systems highlights the need for transparency, reduced waiting time, and multi-device accessibility.

7. Best Practices and Recommendations:

Literature recommends implementing role-based access control, secure authentication, optimized database design, and real-time order tracking. Additionally, structured dashboards and automated workflows are suggested to improve system efficiency and usability.

8. Conclusion and Insights:

The literature review highlights the importance of digital transformation in canteen management systems. These insights guide the development of FlashEats, ensuring it is secure, scalable, user-friendly, and aligned with modern web application standards.

3. REQUIREMENTS DOCUMENT

1. Project Overview

FlashEats is a digital canteen system that replaces manual ordering with an online platform where users can browse menus, place orders, and track status, improving speed and efficiency.

The system targets students and canteen staff, enabling quick food ordering and efficient order management. It reduces delays, improves communication, and ensures smooth and organized canteen operations digitally.

2. Functional Requirements

Users can register, log in, browse menus, add items to cart, place orders, and track status in real time through a secure and responsive platform.

Administrators manage menu items, update orders, and monitor system activity. Role-based access, JWT authentication, and SendGrid OTP verification ensure secure and controlled system access for all users.

3. Technical Requirements

The system uses MERN stack with React, Node.js, Express, and MongoDB, ensuring scalability, fast performance, and smooth interaction between frontend and backend components.

JWT handles authentication while SendGrid manages OTP verification. GitHub manages version control, and AWS EC2 and S3 handle deployment and storage for scalability and reliability.

4. User Experience Requirements

The system provides a simple and intuitive interface allowing users to browse menus and place orders easily with minimal steps and clear navigation.

It is fully responsive across devices with fast loading speed, real-time updates, and smooth interaction, ensuring a seamless experience for both students and administrators.

5. Feasibility Analysis

The system is technically feasible using widely supported MERN technologies with a modular structure that allows easy updates and future enhancements without major changes.

It is economically efficient due to open-source tools and scalable cloud services. Operationally, it improves existing canteen workflows by replacing manual processes with automation.

6. Risks and Challenges

Potential issues include high server load during peak hours, database performance limitations, and security vulnerabilities in authentication and API handling if not properly optimized.

These risks can be reduced through secure coding, optimized APIs, cloud scalability, and continuous testing, monitoring, and system updates for stability and performance.

7. Stakeholder Analysis

Stakeholders include students, administrators, and developers. Students use the system for ordering, while admins manage menus and orders, and developers maintain system functionality.

All stakeholders expect a fast, secure, and transparent system that reduces manual work and improves communication, efficiency, and overall user experience in canteen operations.

8. Conclusion and Recommendations

FlashEats modernizes canteen operations by replacing manual processes with a fast, secure, and scalable digital platform that improves efficiency and user satisfaction.

Future enhancements may include mobile apps, notifications, and analytics. Continuous improvements and security updates will ensure long-term reliability and better system performance.

4. IMPLEMENTATION

1. Empowering Students and Administrators

The system empowers both students and administrators by providing a digital platform for complaint handling. Students can raise issues easily, while administrators efficiently manage and resolve them.

2. Streamlining the Complaint Process

The system streamlines complaint handling by digitizing submission, tracking, and resolution processes. It reduces manual work, minimizes delays, and ensures complaints are managed in an organized and timely manner.

3. Tailored and Role-Based Functionality

The system provides role-based functionality where students can submit and track complaints, while administrators manage resolution. This ensures efficient access control and optimized usage based on user responsibilities.

4. Enhanced User Experience

A clean and responsive interface ensures smooth interaction for all users. Features like real-time updates, simple forms, and intuitive navigation improve usability and overall user satisfaction significantly.

5. Consistency and Transparency

The system ensures consistent complaint handling with standardized formats and clear status updates. This improves transparency between students and administrators while enhancing accountability and reducing confusion in communication.

6. Technological Proficiency and Modernization

The implementation using the MERN stack demonstrates modern web development practices. It integrates frontend, backend, database, and authentication systems into a scalable and efficient application architecture.

7. Adaptability and Accessibility

The platform is fully responsive, allowing access across mobile, tablet, and desktop devices. This ensures users can submit and manage complaints anytime, improving accessibility and convenience.

8. Continuous Improvement

The system can be continuously improved based on user feedback. Future enhancements may include notifications, analytics, and mobile applications to further improve performance, usability, and overall system effectiveness.

4.1 Technologies and tools used in project

The architecture of FlashEats follows a modular MERN stack structure, separating frontend, backend, and database layers. This design ensures scalability, maintainability, and efficient communication between different system components.

React.js (Frontend Framework)

Serves as the foundation for building dynamic and interactive user interfaces.

Used to create dashboards, complaint forms, and real-time status updates for a smooth user experience.

HTML & CSS (Structure and Styling)

HTML structures the layout, forms, and content of the application.

CSS along with Tailwind enhances design, responsiveness, and visual consistency.

JavaScript (Core Scripting Language)

Handles client-side logic, form validation, and interactive features.

Ensures smooth navigation and dynamic updates across the application.

Node.js (Backend Runtime Environment)

Provides the server-side environment to handle requests and authentication.

Enables efficient communication between frontend and database systems.

Express.js (Backend Framework)

Used to build RESTful APIs for complaint handling and user management.

Simplifies routing, middleware, and server-side operations.

MongoDB (Database)

Acts as the primary database for storing user and complaint data.

Provides flexibility with its NoSQL structure and supports scalability.

JWT (Authentication Mechanism)

Used for secure user authentication and session management.

Ensures protected access to routes and system resources.

Nodemailer with SMTP (Email Service)

Used to send OTPs for email verification during registration.

Ensures secure and reliable communication with users.

AWS EC2 (Cloud Hosting Service)

Hosts the backend server in a scalable and reliable environment.

Ensures high availability and performance of the application.

AWS S3 (Storage Service)

Used for storing static files and assets.

Provides secure and efficient cloud-based storage.

Version Control Systems (Git & GitHub)

Git is used for version control and tracking code changes.

GitHub provides repository hosting and supports team collaboration.

Development and Debugging Tools

Browser developer tools are used for debugging and performance monitoring.
Tools like ESLint and Prettier ensure clean and consistent code quality.

Responsive Design and Testing Tools

Used to test application performance across different screen sizes and devices.
Ensures compatibility on mobile, tablet, and desktop platforms.

Documentation Tools

Used to document APIs, project structure, and development processes.
Tools like Postman assist in API testing and validation.

Code Editor (Visual Studio Code)

Serves as the primary development environment with rich extensions.
Provides debugging support, syntax highlighting, and efficient workflow management.

4.2 Architecture of FlashEats Online Canteen Management System

Client-Side Architecture (Frontend Design):

- Uses HTML, CSS, JavaScript, React.js, Tailwind CSS
- HTML defines structure (login, menu, cart, dashboard)
- Tailwind CSS ensures responsive and modern UI design
- React.js handles dynamic rendering and state management
- Enables real-time updates (cart changes, order tracking)
- Component-based structure (menu cards, forms, dashboards)
- Client-side validation ensures correct user input
- Error handling improves user experience

Server-Side Architecture (Backend Development):

- Built using Node.js and Express.js
- Handles request processing and business logic
- Manages authentication, menu, cart, and orders
- Uses RESTful APIs for frontend-backend communication
- Implements JWT-based authentication for security
- Role-based access for users and admins
- Uses SendGrid for email verification
- Handles order processing and status updates
- Ensures secure and efficient data flow

Database Management Architecture:

- Uses MongoDB (NoSQL database)
- Stores data in collections (Users, Menu Items, Orders)
- Flexible schema allows easy updates and scalability
- Stores user details, menu data, and order information
- Uses unique IDs for each record (userId, itemId, orderId)
- Supports fast data retrieval and efficient queries
- Backend performs CRUD operations (Create, Read, Update, Delete)
- Ensures data consistency and integrity
- Enables real-time order tracking and updates
- Scalable for handling large number of users and orders

4.3 Core functionalities of the app

User Functionalities (Ordering System):

- Menu Browsing: Users can view food items with details like name, category, and price
- Cart Management: Add, update, or remove items from cart
- Order Placement: Users can place orders easily through a simple interface
- Order Tracking: Track real-time status (pending, preparing, ready)
- User Authentication: Secure login and registration system
- Communication Transparency: Users stay updated about order progress

Admin Functionalities (Management System):

- Menu Management: Add, update, or delete food items
- Order Monitoring: View all incoming orders in the system
- Order Processing: Update order status (accepted, preparing, completed)
- Filtering and Sorting: Manage orders based on status or time
- User Management: Monitor and control user activities

Form Validation and Error Handling:

- Ensures correct input using client-side validation
- Displays helpful error messages for users
- Improves data accuracy and system reliability

User Interface and Navigation:

- Simple and intuitive interface design
- Easy navigation across pages and features
- Improves overall user experience

System Architecture and Security:

- Secure data handling using MongoDB
- JWT-based authentication for protected access
- Role-based authorization (User/Admin)
- Email verification using SendGrid

Code Structure and Maintainability:

- Clean and well-structured codebase
- Proper documentation for easy understanding
- Supports future updates and scalability

5. RESULT

The core functionalities of the Online Canteen Management System (FlashEats) demonstrate the effectiveness of a centralized platform for managing food ordering efficiently. The system provides a structured and user-friendly environment for users to browse menu items, place orders, and track status, while administrators can manage and process orders in a timely manner.

Order Management and System Functionality:

- Provides structured workflow for food ordering (browse → cart → order → status)
- Enables users to place orders easily with item details and quantity
- Allows real-time order tracking (pending, preparing, ready)
- Admin can manage and update order status efficiently

User Experience and Interface:

- Clean, simple, and user-friendly interface
- Easy navigation across menu, cart, and dashboard
- Fully responsive design for mobile, tablet, and desktop
- Improves overall user interaction and convenience

System Performance and Reliability:

- Fast request handling using optimized backend APIs
- Smooth communication between frontend and backend
- Reliable performance with cloud deployment (AWS EC2 & S3)
- Handles multiple users and orders efficiently

Data Management and Storage:

- Uses MongoDB for efficient data storage
- Stores user details, menu items, and orders securely
- Maintains structured and consistent data
- Enables quick data retrieval and updates

Admin Control and Management:

- Allows administrators to manage menu items easily
- Provides order monitoring and processing system
- Supports filtering and sorting of orders
- Helps prioritize and handle orders efficiently

Authentication and Security Features:

- Implements JWT-based authentication for secure login
- Role-based access for users and admin
- Email verification using SendGrid
- Ensures secure access to system resources

Code Structure and Maintainability:

- Uses MERN stack for modular and scalable architecture
- Maintains clean and organized codebase
- Supports easy updates and future enhancements
- Well-documented structure for better maintenance

5.1 Home Page

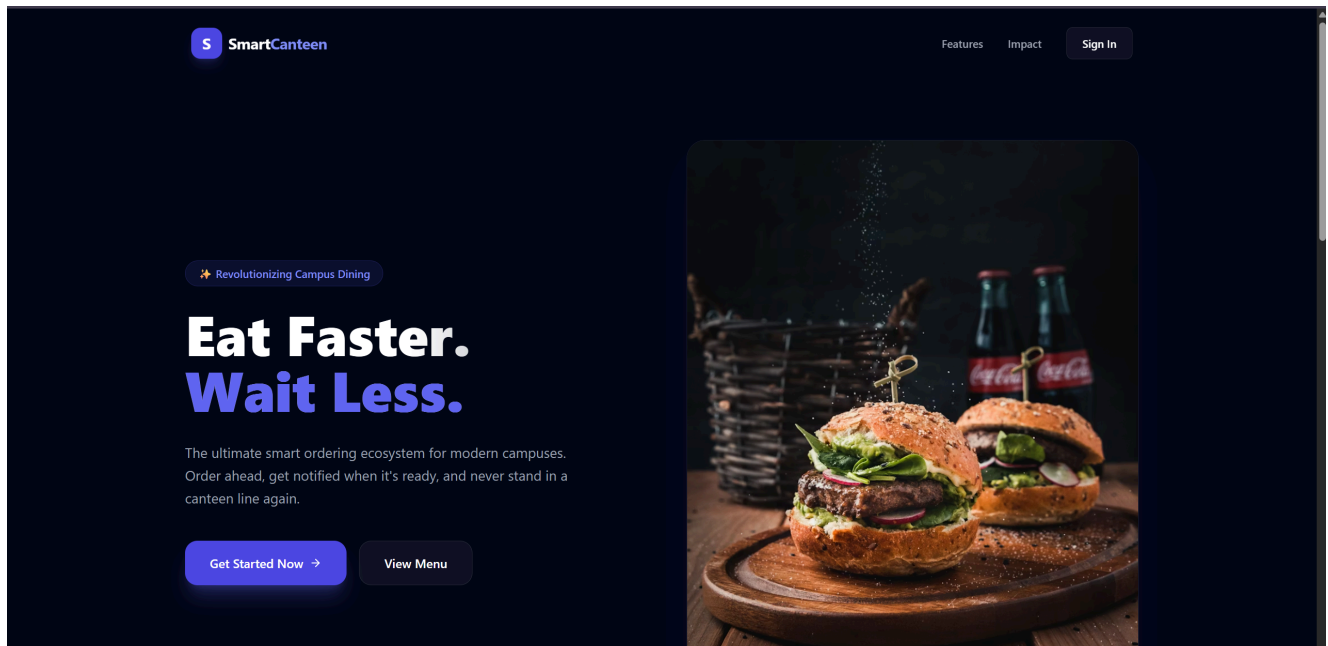


Fig. 5.1.1 Landing Pagewith login and registration

5.2 StudentDashboard

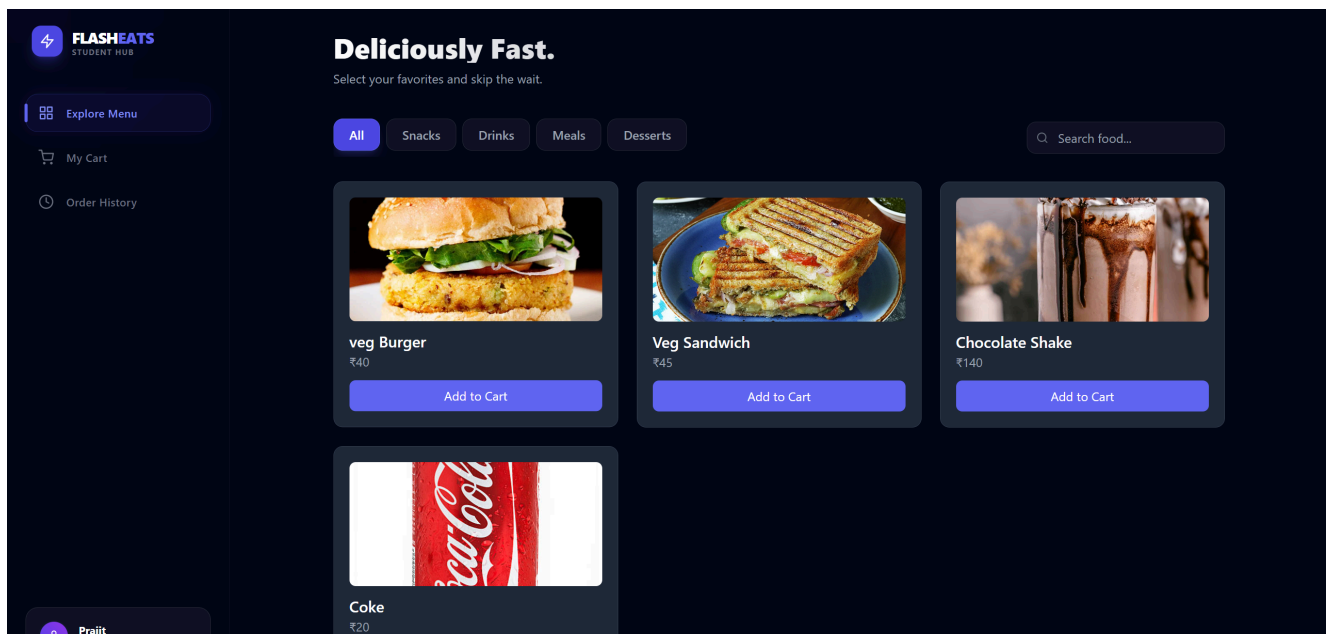


Fig. 5.2.1 Explore Menu Page

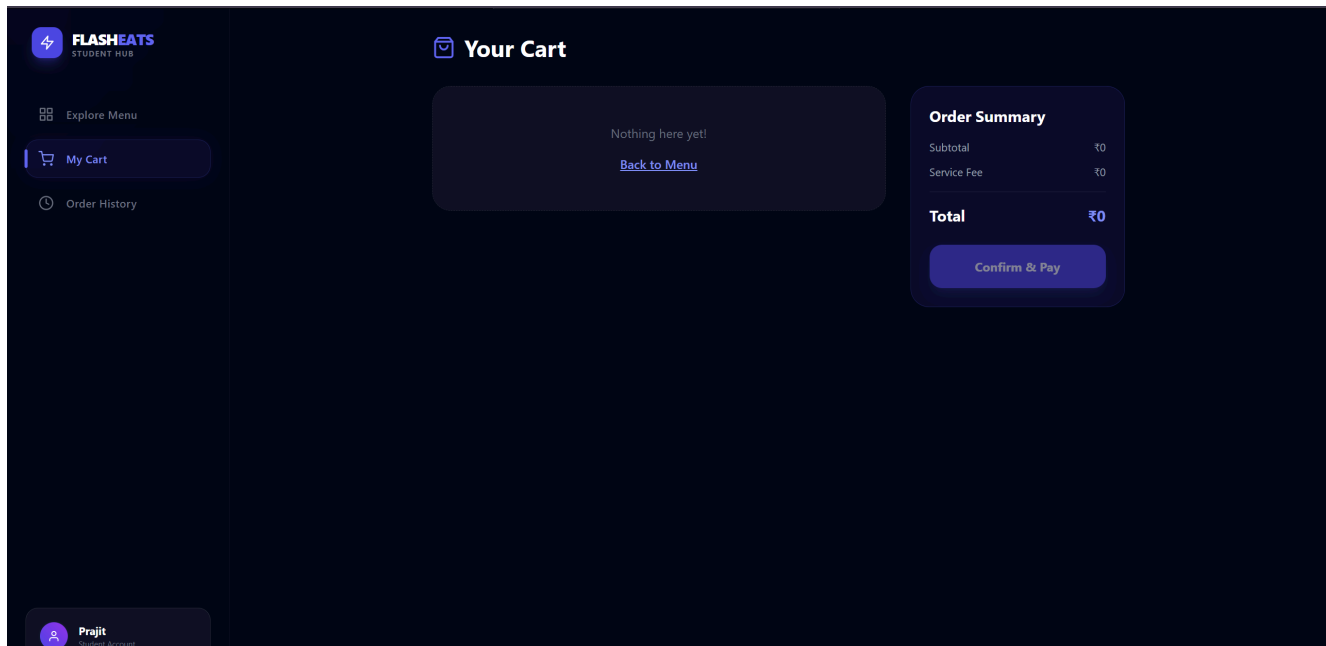


Fig. 5.2.2 Cart Page

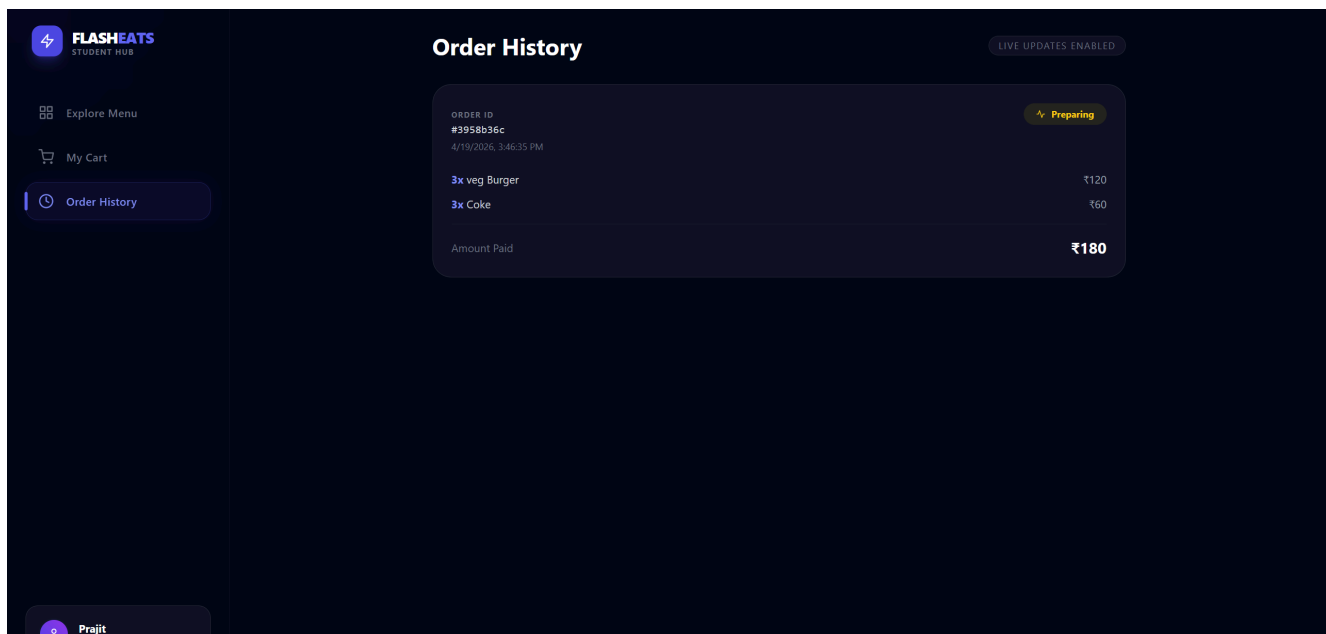


Fig. 5.2.3 Order History Page

5.3 Admin Dashboard

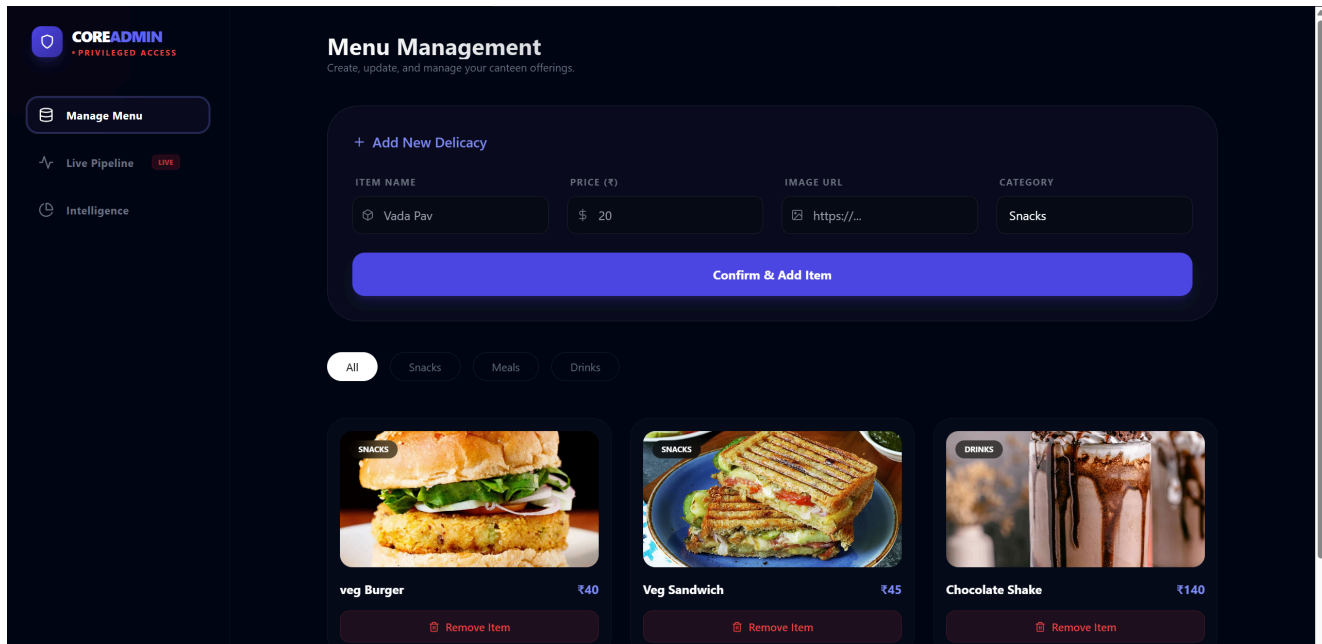


Fig. 5.3.1 Manage Menu Page

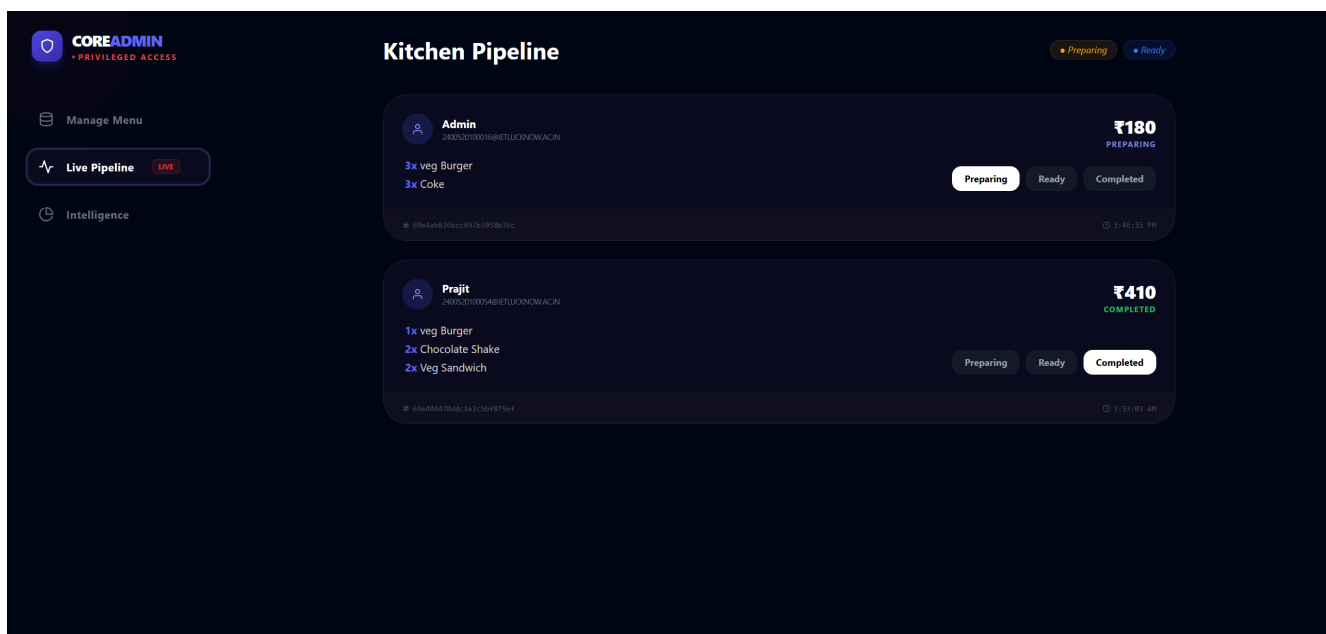


Fig. 5.3.2 Order Pipeline Page

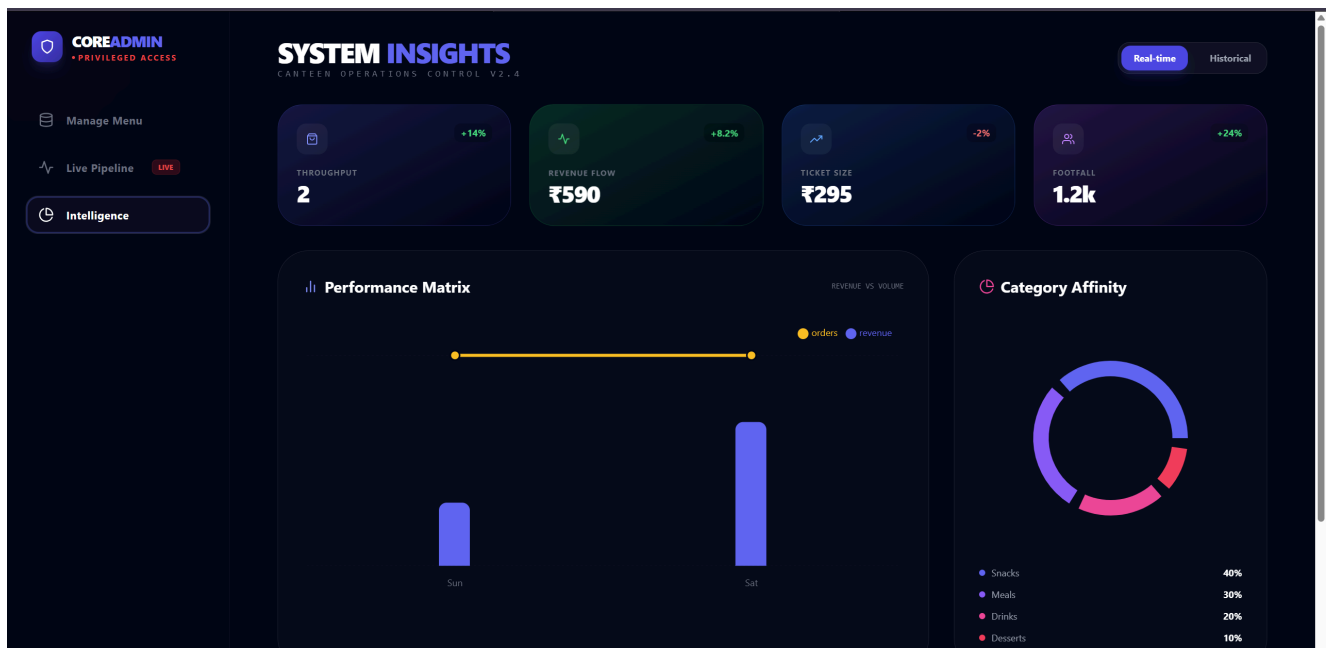


Fig. 5.3.3 Analytics Page

5.4 Database

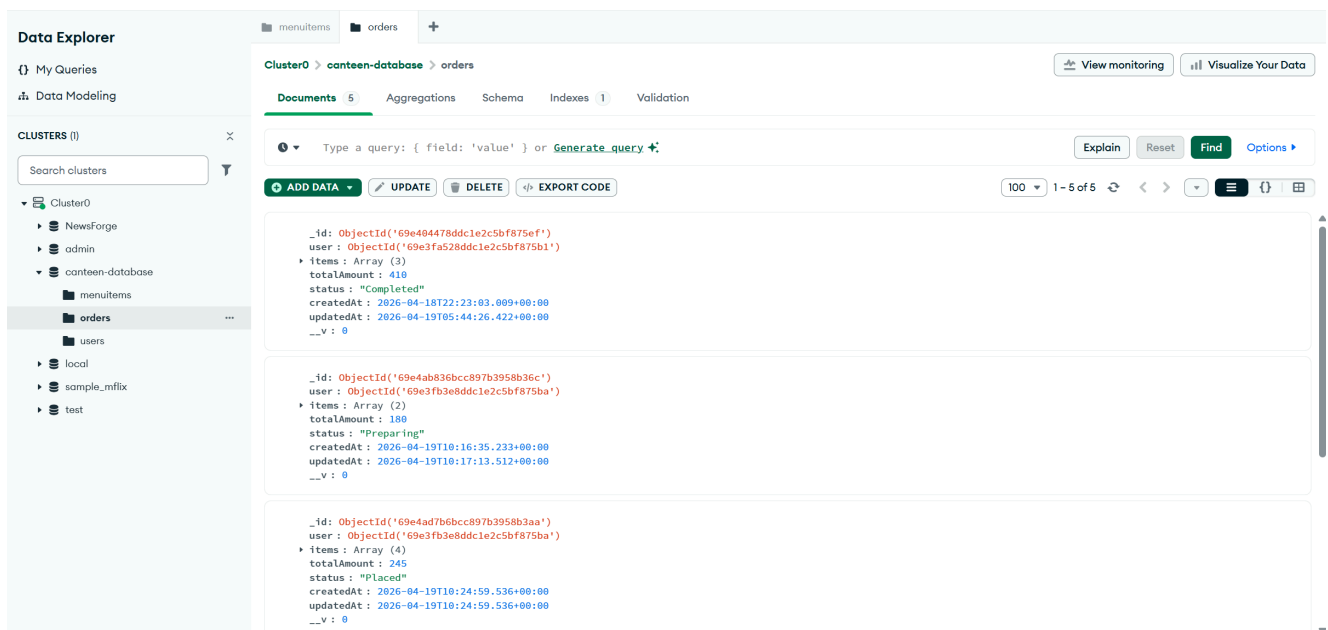


Fig. 5.4.1 Order Database

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - NewsForge
 - admin
 - canteen-database
 - menuitems**
 - orders
 - users
 - local
 - sample_mflix
 - test

Cluster0 > canteen-database > menuitems

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

100 1 - 5 of 5

```

_id: ObjectId('69e401968ddc1e2c5bf875d5')
name: "Veg Burger"
price: 48
image: "https://www.vegrecipesofindia.com/wp-content/uploads/2020/12/burger-re-"
category: "Snacks"
isAvailable: true
createdAt: 2026-04-18T22:11:34.871+00:00
updatedAt: 2026-04-18T22:11:34.871+00:00
__v: 0

_id: ObjectId('69e4035c8ddc1e2c5bf875db')
name: "Veg Sandwich"
price: 45
image: "https://www.indianhealthyrecipes.com/wp-content/uploads/2022/03/veg-gr-"
category: "Snacks"
isAvailable: true
createdAt: 2026-04-18T22:19:08.283+00:00
updatedAt: 2026-04-18T22:19:08.283+00:00
__v: 0

_id: ObjectId('69e4038a8ddc1e2c5bf875de')
name: "Chocolate Shake"
price: 140
image: "https://cookilicious.com/wp-content/uploads/2025/01/Brownie-Milkshake-"
category: "Drinks"
isAvailable: true
createdAt: 2026-04-18T22:19:54.870+00:00

```

Fig. 5.4.2 Menu Database

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - NewsForge
 - admin
 - canteen-database
 - menuitems
 - orders
 - users**
 - local
 - sample_mflix
 - test

Cluster0 > canteen-database > users

Documents 4 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#)

100 1 - 4 of 4

```

_id: ObjectId('69e3fa528ddc1e2c5bf875b1')
name: "Prajit"
email: "2400520100054@ietlucknow.ac.in"
password: "$2b$10$9YfA7r7pXg/b8IammiZmWIEPT6QFTQRQ./zsoIQ1lc.bhIKUKV6"
role: "student"
isVerified: false
createdAt: 2026-04-18T21:40:34.196+00:00
updatedAt: 2026-04-18T21:40:34.196+00:00
__v: 0

_id: ObjectId('69e3fb3e8ddc1e2c5bf875ba')
name: "Admin"
email: "2400520100016@ietlucknow.ac.in"
role: "admin"
isVerified: false
otp: null
otpExpiry: null
createdAt: 2026-04-18T21:44:30.003+00:00
updatedAt: 2026-04-20T10:18:49.123+00:00
__v: 0

_id: ObjectId('69e3fc4b8ddc1e2c5bf875c4')
name: "Admin"
email: "2400520200008@ietlucknow.ac.in"
role: "admin"
isVerified: false
otp: null

```

Fig. 5.4.3 User Database

6. FUTURE ENHANCEMENTS

Key Enhancements:

- **Payment Gateway Integration:** Add support for UPI, cards, and digital payments for seamless transactions
- **Mobile Application:** Develop Android and iOS apps for better accessibility and convenience
- **Delivery Tracking System:** Enable real-time tracking of order status from preparation to delivery
- **Feedback and Rating System:** Allow users to rate food items and provide feedback for service improvement

Enhanced Management and Analytics:

- Analytics dashboard for tracking popular items and order trends
- Helps administrators make better business decisions

Technical and Scalability Improvements:

- Cloud optimization for better performance and reliability
- Microservices architecture for easier scaling and maintenance

Security Enhancements:

- Advanced authentication mechanisms
- Regular security updates to protect user data

Advanced Communication:

- Email notifications using SendGrid
- In-app notifications for real-time updates

7. REFERENCES

The following references were utilized for the design, implementation, and foundational knowledge of the Online Canteen Management System (FlashEats):

1. **MongoDB Documentation**
<https://www.mongodb.com/docs/>
(Used for database design and data modeling)
2. **Express.js Official Documentation**
<https://expressjs.com/>
(Used for backend API development)
3. **React.js Documentation**
<https://react.dev/>
(Used for building the frontend user interface)
4. **Node.js Documentation**
<https://nodejs.org/en/docs/>
(Used for server-side development)
5. **JSON Web Token (JWT) Documentation**
<https://jwt.io/introduction>
(Used for authentication and role-based access control)
6. **SendGrid Documentation**
<https://docs.sendgrid.com/>
(Used for sending email verification and notifications)
7. **AWS EC2 Documentation**
<https://docs.aws.amazon.com/ec2/>
(Used for backend deployment)
8. **AWS S3 Documentation**
<https://docs.aws.amazon.com/s3/>
(Used for file storage and static asset hosting)
9. **GitHub Documentation**
<https://docs.github.com/>
(Used for version control and collaboration)
10. **Stack Overflow**
<https://stackoverflow.com/>
(Used for troubleshooting and resolving development issues)
11. **MDN Web Docs**
<https://developer.mozilla.org/>
(Used for HTML, CSS, and JavaScript references)